

This is due by July 18.

Problem 1. Identities and inverses are special elements in groups and rings. They are unique if they exist. So there is no ambiguities/choices when we write “Let $e \in G$ be *the* identity element.” Let G be a group and $g \in G$.

1. Suppose $e, e' \in G$ both are identities. Then $e = e'$.
2. Suppose g^{-1}, h both are inverses of g . Then $g^{-1} = h$.
3. (Just think about this. You don't need to write it down.) Let R be a ring. Show the additive identity $0 \in R$ and the additive inverse $-r$ of any ring elements $r \in R$ are unique in R .

Problem 2. Between sets, there are functions connecting them. Between groups, there are special functions (functions that satisfy extra properties) connecting them. Let G, H be groups. A function $f : G \rightarrow H$ is a group homomorphism if $f(gg') = f(g)f(g')$ for all $g, g' \in G$. That is, a group homomorphism is a function that preserves group operation.

1. Suppose $f : G \rightarrow H$ is a group homomorphism. Show that
 - (a) $f(e_G) = e_H$. *That is, a group homomorphism preserves identity element.*
 - (b) $f(g^{-1}) = (f(g))^{-1}$. *That is, a group homomorphism preserves inverses.*
2. Let $m \geq 0$. Consider the function $\rho : \mathbb{Z} \rightarrow \mathbb{Z}_m$ by $\rho(k) = [k]$. Show ρ is a group homomorphism. A group homomorphism is called a group isomorphism if as a function it is bijective. Show $\rho : \mathbb{Z} \rightarrow \mathbb{Z}_m$ is not a group isomorphism.
3. We usually write $G \cong H$ if there is a group isomorphism $f : G \rightarrow H$. Show $2\mathbb{Z} \cong 3\mathbb{Z}$ as subgroups of $(\mathbb{Z}, 0, +)$.

proof of part 3. The usual way is to define $\rho : 2\mathbb{Z} \rightarrow 3\mathbb{Z}$ by $2k \mapsto 3k$ and show ρ is an bijective group homomorphism.

A slight variation is to construct the inverse homomorphism $\phi : 3\mathbb{Z} \rightarrow 2\mathbb{Z}$ by $3k \mapsto 2k$ and note that $\phi\rho = \text{id}$ and $\rho\phi = \text{id}$. This implies that both ρ and ϕ are injective and surjective. Hence both are bijective.

Another way to use Problem 3 part 2 where we showed that, as they are cyclic groups of infinite order, $2\mathbb{Z}$ and $3\mathbb{Z}$ both isomorphic to $\mathbb{Z}$. Hence $2\mathbb{Z} \cong \mathbb{Z} \cong 3\mathbb{Z}$. □

Problem 3. We investigate the group structure of cyclic groups. Let G be a group and $g \in G$ be an element. Recall the cyclic group generated by g is $\langle g \rangle = \{g^i : i \in \mathbb{Z}\}$.

1. Show the followings are all equivalent.

- (a) $\langle g \rangle$ is of finite order.
- (b) There exists some integers $i \neq j$ such that $g^i = g^j$.
- (c) there exists some $N \in \mathbb{N}$ such that $g^N = e$.

Then conclude that $\langle g \rangle$ is of infinite order iff $g^i \neq g^j$ for all integers $i \neq j$.

- 2. Show that $\langle g \rangle$ is of infinite order iff $\langle g \rangle \cong \mathbb{Z}$.
- 3. Show that $\langle g \rangle$ is of finite order n iff $\langle g \rangle \cong \mathbb{Z}_n$.
- 4. Show that any subgroup $H \subseteq \langle g \rangle$ is a cyclic group generated by g^k where

$$k = \min\{i > 0 : g^i \in H\}.$$

- 5. The subgroup that contains only the identity of any group is called the trivial subgroup. Suppose $\langle g \rangle$ is of infinite order. Conclude that all nontrivial subgroups of $\langle g \rangle$ is of infinite order and thus is isomorphic to $\mathbb{Z}$.
- 6. (*Just think about this. You don't need to write it down.*) Compare part 5 with Problem 2 part 4. Do you have a different proof for that question?

Proof. 1. Suppose $\langle g \rangle$ is of finite order. Consider the infinite list of natural-number-powers of g , say $\{e = g^0, g^1, g^2, \dots\}$. There must be repetition somewhere. That is, there exists natural numbers $i \neq j$ such that $g^i = g^j$. WLOG, assume $j > i$. Then there exists $N = j - i$ such that $g^N = g^j g^{-i} = g^j g^{-j} = e$. This shows (a) $\implies$ (b) $\implies$ (c). It remains to show (c) $\implies$ (a). (Delano.) Suppose $g^N = e$ for some $N \in \mathbb{N}$. Then by the division algorithm for all $n \in \mathbb{Z}$, $g^n = g^{qN+r} = g^r$ for some $0 \leq r < N$. This means $\langle g \rangle = \{e = g^0, g^1, \dots, g^{N-1}\}$.

Negating statements (a) and (b) gives the desired result.

- 2. Consider $\rho : \mathbb{Z} \rightarrow \langle g \rangle$ by $n \mapsto g^n$. One can show this is a surjective group homomorphism. The kernel $\ker \rho = \{d : g^d = e\}$. Recall the kernel is a subgroup and hence $\ker \rho = (N)$ for the least natural number N such that $g^N = e$. Also note part 1 (a) and (c) and the deduction on $\langle g \rangle$ of infinite order. Now $\langle g \rangle$ is of infinite order iff $\ker \rho = 0$ iff $N = 0$. In this case, $\langle g \rangle \cong \mathbb{Z}$. And $\langle g \rangle$ is of finite order iff $\ker \rho = (N)$ for some $N > 0$. In this case, $\langle g \rangle \cong \mathbb{Z}_N$. We have proved both (2) and (3).
- 3. (4) is essentially the division algorithm. (Jenny, Marcial, and Tianshuo.) Let $g^n \in H$ and consider $n = qk + r$ for $0 \leq r < k$. We show $r = 0$. Suppose $r > 0$. Then $g^r = g^n ((g^k)^{-1})^q \in H$ which contradicts the minimality of k . $\square$

Problem 4. We continue to investigate the structure of cyclic group $\langle g \rangle$ when it is of finite order. Suppose that $\langle g \rangle$ is of finite order n .

1. Show g is of order n iff $\langle g \rangle$ is of order n . Conclude $\langle g \rangle = \{e, g, g^2, \dots, g^{n-1}\}$.
2. Let $0 \leq i < n$. Suppose $(g^i)^k = e$. Show $n \mid ik$.
3. Let $0 \leq i < n$. Show g^i is of order $\frac{n}{(n,i)}$.
4. Let $0 \leq i, j < n$. Suppose g^i and g^j both are of order d . Then $g^j = (g^i)^k$ for some $k \in \mathbb{Z}$.
5. Let $d \mid n$ be positive. Then $H \subseteq G$ is a subgroup of order d iff

$$H = \{g^k : 0 \leq k < n \text{ and } \frac{n}{(n,k)} = d\}.$$

6. Conclude that in a finite cyclic group of order n ,
 - (a) there is only one subgroup H_d of order d for each $d \mid n$
 - (b) and that subgroup H_d is the cyclic group generated by g^k where $\frac{n}{(n,k)} = d$.
7. (Just think about this. You don't need to write it down.) Compare part 6 with Problem 2 part 4 and see how the subgroups of $\mathbb{Z}$ is different from that of a cyclic group of finite order.

Proof. 1. Write the $\langle g \rangle = \{e = g^0, g, g^2, \dots\}$. Note that g is of order n iff the N in the proof of Problem 3 part 2 equals n . Then g is of order n iff $\langle g \rangle \cong \mathbb{Z}_n$ which is of finite order.

2. (2) is also the division algorithm. Consider $ik = qn + r$ for some $0 \leq r < n$. Suppose $r > 0$. Since $g^n = e$, we have $g^{ik} = g^r$. But $g^r \neq e$ otherwise g has order at most $r < n$.
3. Let the order of g^i be N . By part 2, we know $n \mid iN$. By Euclid's lemma, $\frac{n}{(n,i)} \mid N$. So we may write $N = t \cdot \frac{n}{(n,i)}$ for some $t \in \mathbb{N}$. Note by assumption $g^n = e$. and clearly $n \mid i \cdot \frac{n}{(n,i)}$ for $\frac{i}{(n,i)} \in \mathbb{Z}$. Recall that N is the least natural number such that $g^N = e$. So $N = \frac{i}{(n,i)}$ since $t = 1$ gives the least possible multiple of $\frac{n}{(n,i)}$.
4. (Fred.) Let $d = \frac{n}{(n,i)}$ and write $g = (n, i)$. So there exists $an + bi = g$ for some $a, b \in \mathbb{Z}$. By part 3, $d = \frac{n}{(n,j)}$. So $(n, j) = (n, i) = g$. Then $gl = j$ for some $l \in \mathbb{Z}$ which implies $anl + bil = gl = j$ for some $a, b \in \mathbb{Z}$. This mean $g^j = g^{anl+bil} = (g^i)^{bl}$.
5. Note that $H = \langle g^t \rangle$ for some t by Problem 3 part 4. By Problem 4 part 3, $\frac{n}{(n,t)} = d$. Let $t = n/d$. By Problem 4 part 4, if g^j is of order d , then $g^j = (g^t)^k$ for some $l \in \mathbb{Z}$. So for each $0 \leq j < n$ such that $\frac{n}{(n,j)} = d$, we know $g^j \in H$. Now it suffices to show the for any $0 \leq k < d$, kt satisfies $\frac{n}{(n,kt)} = d$. Indeed, this is true because $\frac{n}{(n,kt)} = \frac{n}{(n, \frac{nk}{d})} = \frac{n}{\frac{1}{d}(dn, nk)} = \frac{nd}{n(d,k)} = d$. $\square$

Problem 5. We prove $\mathbb{Z}[\sqrt{2}]$ is a domain in another way.

1. Show a subring of a field F is a domain.
2. Show $\mathbb{Q}[\sqrt{2}] = \{a + b\sqrt{2} : a, b \in \mathbb{Q}\}$ is a field.
3. Deduce that $\mathbb{Z}[\sqrt{2}]$ is a domain.